

Draft Framework: Ai+Ethics, Emergent Intelligence, and Experiential Convergence

Potential Collaboration with Syracuse University's BioInspired Institute and EmIRGE-Bio Program

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Overview

Following our recent email exchange regarding a potential guest lecturer role within Syracuse University's BioInspired Institute and the NSF-supported *Emergent Intelligence Research for Graduate Excellence in Biological and Bio-Inspired Systems* (EmIRGE-Bio) program, I wanted to synthesize several of the ideas into a more structured framework for continued conversation and collaborative development.

This document is intended less as a formal proposal and more as a potential action framework outlining how a living-systems approach to *Ai+Ethics* (Ai = augmented intelligence), participatory evaluation, experiential learning, and convergence research might be integrated within the evolving EmIRGE-Bio ecosystem.

In particular, the framework builds directly upon themes articulated within:

- the *NRT Advisory Board: Recommendations and Considerations*,
and
- the *Strategic Briefing: Enhancing Convergence through Integrated Experiential Learning*.

These documents collectively emphasize:

- Team Science,
- emergent intelligence,
- experiential learning,
- AI ethics,
- convergence research,
- student-led ecosystems,
- and interdisciplinary collaboration.

The framework below attempts to operationalize these objectives through:

1. development of a living-systems Ai+Ethics curriculum,
2. integration of the Story of the Plate experiential platform,
3. launch of an interdisciplinary Emergence-Lab (eLab),
4. and the continued development of participatory and governance-aware approaches to AI within biological and socio-ecological systems.

More broadly, I believe this collaboration could position Syracuse University as an important co-design partner in the ongoing development of the *Resilient Community Index (RCI)* framework (outlined in draft research paper I sent) and its broader convergence research agenda surrounding biodiversity, climate resilience, governance-aware AI systems, and participatory ecological evaluation.

I. Living-Systems Approach to Ai+Ethics

Alignment with EmIRGE-Bio

The proposed framework directly supports the Advisory Board recommendation to integrate AI ethics into the *BioInspired Communication and Ethics* course through a new instructional module focused on responsible and transparent AI use, peer review, and ethical decision-making in research environments.

Building upon this recommendation, the proposed Ai+Ethics framework introduces a living-systems approach to artificial intelligence grounded in:

- emergent intelligence,
- biological organization,
- relational systems,
- sustainability science,
- participatory evaluation,
- and governance-aware AI systems.

Within this framework, AI is approached not primarily as autonomous computation, but as a form of *augmented intelligence* embedded within ecological, biological, and human systems.

The modules would also introduce the complementary concept of *formative immergence*, which extends conventional theories of emergence by exploring how relational organization within living systems may involve recursive and potentially quantum-informed dimensions of coherence not reducible to bottom-up computational aggregation alone.¹

¹ A 2024 study entitled “Ultraviolet Superradiance from Mega-Networks of Tryptophan in Biological Architectures” co-authored by N.S. Babcock found that mega-networks of tryptophan molecules in brain microtubules can exhibit quantum superradiance, a phenomenon where they collectively emit ultraviolet

Drawing from systems biology, relational realism, and emerging work in quantum biology, the framework encourages students to critically examine:

- the distinction between computational simulation and biological sentience found in living systems,
- the role of material substrate in cognition,
- and the ethical implications of applying AI within biological systems.

This direction aligns closely with the Advisory Board document’s call to critically examine computational functionalism and the “challenge of substrate and agency” within AI-enabled biological research environments.

II. Story of the Plate as an Experiential Anchor

Consistent with the Strategic Briefing’s recommendation to formalize an “Experiential Anchor” model within the EmIRGE-Bio curriculum, the proposed Ai+Ethics modules would become a core component of the *Story of the Plate* platform—an open-architecture curriculum integrating:

- ecological monitoring,
- garden-based learning,
- participatory sustainability assessment,
- circular-economy systems,
- and governance-aware AI tools.

The platform would function as a student-led experiential ecosystem connecting:

- soil health,
- nutrient density,
- microbiome dynamics,
- food systems,
- AI-assisted ecological monitoring,
- and socio-ecological resilience.

Through experiential participation, students would investigate how ecological systems, biological intelligence, and governance-aware computational systems interact across multiple scales.

light faster and more strongly than acting individually. Published in The Journal of Physical Chemistry B, this research suggests that these, in turn, can be used for things like quantum computing in the brain, as well as signaling and protecting the brain against degenerative diseases

In this way, AI ethics becomes less an abstract compliance exercise and more a living-systems question embedded within ecological and social realities.

This framework also aligns with the Strategic Briefing's emphasis on integrating "Garden-Based Learning" with STEM through the proposed Emergence-Lab (eLab) ecosystem.

III. Emergence-Lab (eLab) Residency Framework

Two-Week Residency Structure

Following preliminary planning and stakeholder coordination over the summer, the collaboration could consider including a two-week interdisciplinary residency early-Fall modeled in part after frameworks such as the Watson Distinguished Lecture series.

The residency would function as both:

- a public intellectual event,
- and a participatory design process supporting convergence research, Team Science, and experiential learning that brings together strategic stakeholders to launch e-Lab including community partners.

A. Public Lecture Series

The residency could include three public lectures centered on themes such as:

1. A Living-Systems Approach to Ai+Ethics
2. Emergent Intelligence, Formative Immersion, and Bio-Inspired AI
3. Participatory Evaluation, Sustainability, and Governance-Aware AI Systems

These lectures would be open to:

- students,
- faculty,
- sustainability practitioners,
- healthcare stakeholders,
- community organizations,
- and regional collaborators.

B. Campus-Wide Design Charette

The residency could also include a two-day interdisciplinary design charette involving:

- faculty across STEM and humanities,
- students,

- sustainability organizations,
- public health groups,
- food systems participants,
- community stakeholders,
- and regional collaborators.

The purpose of the charette would be to collaboratively explore:

- the structure of an Emergence-Lab (eLab),
- implementation pathways for the Story of the Plate platform,
- convergence-oriented curricular initiatives,
- and participatory governance models for AI-enabled socio-ecological systems.

The design charette would model the Team Science approach emphasized throughout the EmIRGE-Bio framework by integrating biological, computational, ecological, ethical, and community perspectives within a shared participatory process.

IV. Preliminary Research and Stakeholder Development

Leading up to the residency, preliminary research and organizational development activities could be conducted to:

- identify institutional and community stakeholders,
- establish collaborative relationships,
- assess opportunities for interdisciplinary integration,
- and prepare a draft action framework to guide the two-day design charette.

Potential stakeholder categories may include:

- faculty across biology, engineering, sustainability, and ethics,
- local food systems organizations,
- regenerative agriculture initiatives,
- K–12 educational partners,
- public health groups,
- and community-based environmental organizations.

This participatory structure aligns directly with the Strategic Briefing’s emphasis on student-led convergence ecosystems integrating service-learning, outreach, and circular-economy systems.

V. Convergence Track and Professional Development Pathways

The proposed eLab initiative could support the Strategic Briefing's broader vision for a continuous convergence track integrating:

- student-led mini-conferences,
- trainee grant programs,
- service-learning,
- interdisciplinary collaboration,
- and industry-facing experiential learning opportunities.

Consistent with the Advisory Board recommendations document, the initiative may also support the development of:

- an "AI Ethics and Responsible Data Science" microcredential,
- faculty mentoring workshops on AI ethics,
- and student-led seminars focused on AI, science policy, sustainability, and governance.

As the Ai+Ethics curriculum matures, future partnership opportunities may also be explored with IEEE and related organizations to support professional certification pathways grounded in:

- governance-aware AI systems,
- biologically informed intelligence,
- participatory evaluation,
- and sustainability-oriented AI ethics.

This direction would complement the Strategic Briefing's emphasis on experiential badging, interdisciplinary professional development, and workforce preparation for the emerging biobased economy.

VI. Integrated Student-Led and Experiential Components

Consistent with the Strategic Briefing's recommendation for a continuous convergence track emerging from the eLab ecosystem, the proposed framework would support a series of integrated student-led and experiential learning components designed to connect AI ethics, living systems research, sustainability, and professional development within real-world environments.

Rather than functioning solely as classroom instruction, the Ai+Ethics and Story of the Plate framework would operate as a scaffolded experiential ecosystem through which students engage

interdisciplinary collaboration, participatory evaluation, and convergence-oriented problem solving.

Component	Potential Strategic Implementation	Real-World Skills Developed
Student-Led Mini-Conference	Students curate interdisciplinary events involving sustainability practitioners, AI researchers, community organizations, and bioeconomy stakeholders connected to the eLab ecosystem.	Scholarly leadership, event coordination, interdisciplinary communication
Trainee Grant Program	Students develop small-scale interdisciplinary project proposals connected to Story of the Plate, participatory ecological monitoring, governance-aware AI systems, or circular-economy initiatives.	Grant writing, budgeting, collaborative research development
Service-Learning & Outreach	Students participate in community-based ecological monitoring, regenerative food systems initiatives, garden-based learning, and sustainability outreach connected to the Story of the Plate platform.	Leadership, mentorship, participatory research, public engagement
Industry Internship Track	Potential partnerships with bioeconomy firms, sustainability labs, AI governance organizations, and regional food systems initiatives connected to Syracuse University's broader convergence ecosystem. ²	Sector-bridging, technology transfer, interdisciplinary workforce preparation
Ai+Ethics Microcredential Pathway	Development of stackable professional development modules aligned with AI ethics, governance-aware AI systems, participatory evaluation, and responsible data science.	Ethical reasoning, AI governance literacy, professional certification readiness
eLab Participatory Design Projects	Students collaborate with faculty and community stakeholders in iterative design, assessment, and refinement of eLab and RCI-related initiatives.	Team Science, systems thinking, participatory design, adaptive evaluation

² For more see the following article that was released on May 21, 2025: "University's Dynamic Sustainability Lab and Ireland's BiOrbic Sign MOU to Advance Markets for the Biobased Economy," Syracuse University Today, accessed March 5, 2026, ([Link](#)).

Together, these components support the Strategic Briefing's broader objective of transitioning students from theoretical learners into real-world leaders capable of navigating the intersections of AI, climate resiliency, biotechnology, sustainability, and participatory governance.

VII. Long-Term Vision and RCI Co-Design Partnership

Beyond the immediate curricular and residency activities, this collaboration could position Syracuse University's BioInspired Institute as a co-design partner in the continued development of the *Resilient Community Index (RCI)* framework.

The RCI integrates:

- ecological indicators,
- participatory evaluation,
- relational goods,
- governance-aware AI systems,
- nutrient density metrics,
- and soil-to-synapse models of resilience

within a unified socio-ecological framework for assessing biodiversity and ecosystem responses to climate change.

The BioInspired Institute's emphasis on emergent intelligence, biological systems, Team Science, and interdisciplinary convergence provides a uniquely compatible environment for advancing these concepts through both theoretical inquiry and experiential implementation.

This collaboration also aligns with Syracuse University's broader efforts to support the emerging biobased economy through initiatives such as the Dynamic Sustainability Lab and the BiOrbic partnership focused on advancing markets for sustainable biological systems (see footnote 2).

VIII. Ongoing Asynchronous Guest Lecturer Role

Following the residency and initial launch activities, the proposed role could continue in an asynchronous guest lecturer capacity throughout the EmIRGE-Bio NSF grant period.

This ongoing role may include:

- continued development and refinement of Ai+Ethics modules,
- virtual guest lectures,
- mentorship of interdisciplinary student projects,
- curriculum consultation,

- collaborative workshops,
- participatory evaluation support,
- and contribution to evolving eLab initiatives.

This structure allows sustained engagement while remaining flexible, scalable, and responsive to the evolving needs of the EmIRGE-Bio program.

Closing Vision

At its core, this collaboration seeks to reposition AI ethics within the continuity of living systems rather than treating intelligence as separable from ecological, biological, and relational conditions.

By integrating augmented intelligence, emergent intelligence, formative immergence, Team Science, participatory design, and experiential ecological learning, the proposed framework aims to create a scalable interdisciplinary environment capable of supporting both scientific innovation and human-centered stewardship in an era of accelerating technological and ecological change.

In doing so, the initiative supports the EmIRGE-Bio program's broader vision of advancing convergence research around the "Understanding the Rules of Life" framework while preparing students to become leaders in AI, climate resiliency, biotechnology, and the emerging biobased economy.